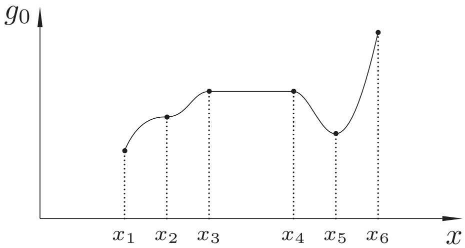
Chapter 3 – Basics of convex programming

3.1 Local and global optima

– General optimization problem $(\mathbb{P})$ under inequality constraints:

$$\begin{aligned} \min_{\mathbf{x}} : \quad & g_0(\mathbf{x}) \\ \text{s.t.} : \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, l \\ & \mathbf{x} \in \mathcal{X} \end{aligned}$$

– Feasible point
  Existence of feasible points
  Feasible set
– Global optimum
  Existence of a global optimum
– Local optimum
– Stationary point



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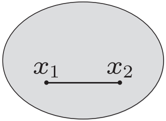
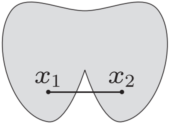
3.2 Convexity

– A set $\mathcal{S} \subset \mathbb{R}^n$ is convex if for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{S}$ and all $\lambda \in (0, 1)$ it holds that:

$$\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in \mathcal{S}$$

– Convex vs. non-convex set:



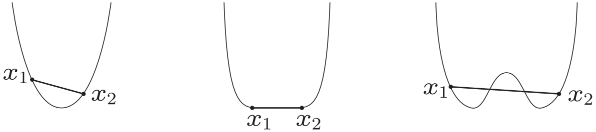


3.2 Convexity

- A function $f : \mathcal{S} \to \mathbb{R}$ is convex (on the convex set $\mathcal{S} \subset \mathbb{R}^n$) if for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{S}$ with $\mathbf{x}_1 \neq \mathbf{x}_2$ and all $\lambda \in (0, 1)$ it holds that:

$$f(\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1-\lambda) f(\mathbf{x}_2)$$

- The function f is strictly convex if strict inequality $(<)$ holds instead.
- Strictly convex, convex, non-convex function:



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3.2 Convexity

- The set $\mathcal{S} = \{\mathbf{x} \in \mathcal{X} : g_i(\mathbf{x}) \leq 0, i = 1, \ldots, l\}$ is convex if the functions $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are convex.
- An optimization problem $(\mathbb{P})$ is convex if and only if the objective function is convex and the feasible set is convex. In that case:
  - → Local optimum $=$ global optimum.
  - → There is not always a solution.
  - → There is not always a single solution.

3.2 Convexity

– Let $f : \mathcal{S} \to \mathbb{R}$, where $\mathcal{S}$ is convex and f is continuously differentiable. Then f is (strictly) convex if and only if the gradient ∇f is (strictly) monotonic.

  A function $\mathbf{g} : \mathcal{S} \to \mathbb{R}^n$ is monotonic on $\mathcal{S}$ if for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{S}$ with $\mathbf{x}_1 \neq \mathbf{x}_2$ it holds that:

$$(\mathbf{x}_2 - \mathbf{x}_1)^T(\mathbf{g}(\mathbf{x}_2) - \mathbf{g}(\mathbf{x}_1)) \geq 0$$

  The function $\mathbf{g}$ is strictly monotonic on $\mathcal{S}$ if strict inequality holds here.

3.2 Convexity

– Let $f : \mathcal{S} \to \mathbb{R}$, where $\mathcal{S}$ is convex and f is twice continuously differentiable. Then

(i) f is convex if and only if the Hessian $\nabla^2 f$ is positive semidefinite.
(ii) f is strictly convex if and only if the Hessian $\nabla^2 f$ is positive definite.

Hessian:

$$\nabla^2 f = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

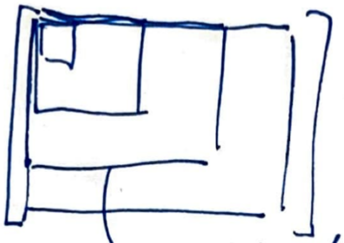
A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive semidefinite if $\mathbf{y}^T \mathbf{A} \mathbf{y} \geq 0 \quad \forall\, \mathbf{y} \in \mathbb{R}^n$

A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive definite if $\mathbf{y}^T \mathbf{A} \mathbf{y} > 0 \quad \forall\, \mathbf{y} \in \mathbb{R}^n$

How to check if a symmetric matrix is positive (semi)definite?

→ Sylvester's criterion
→ positive definite if a Cholesky decomposition exists
→ positive definite if all eigenvalues are strictly positive

→ Sylvester's criterion :



det of all these matrices is positive

→ Cholesky decomposition :

all diag $\neq 0$: pos def

some diag $= 0$: pos semidef

$A = LL^T$, where L is a non-singular lower triangular matrix.

→ Eigen decomposition

$\lambda_i > 0$: pos def

$A = V \Lambda V^T$ with $\lambda_i > 0$. $\lambda_i \geq 0$: pos semidef

3.2 Convexity

- Let $\mathcal{S}$ be a convex set. If $f : \mathcal{S} \to \mathbb{R}$ and $g : \mathcal{S} \to \mathbb{R}$ are convex and $h : \mathcal{S} \to \mathbb{R}$ is strictly convex, then:
  - → αf is convex
  - → $f + g$ is convex
  - → $f + h$ is strictly convex

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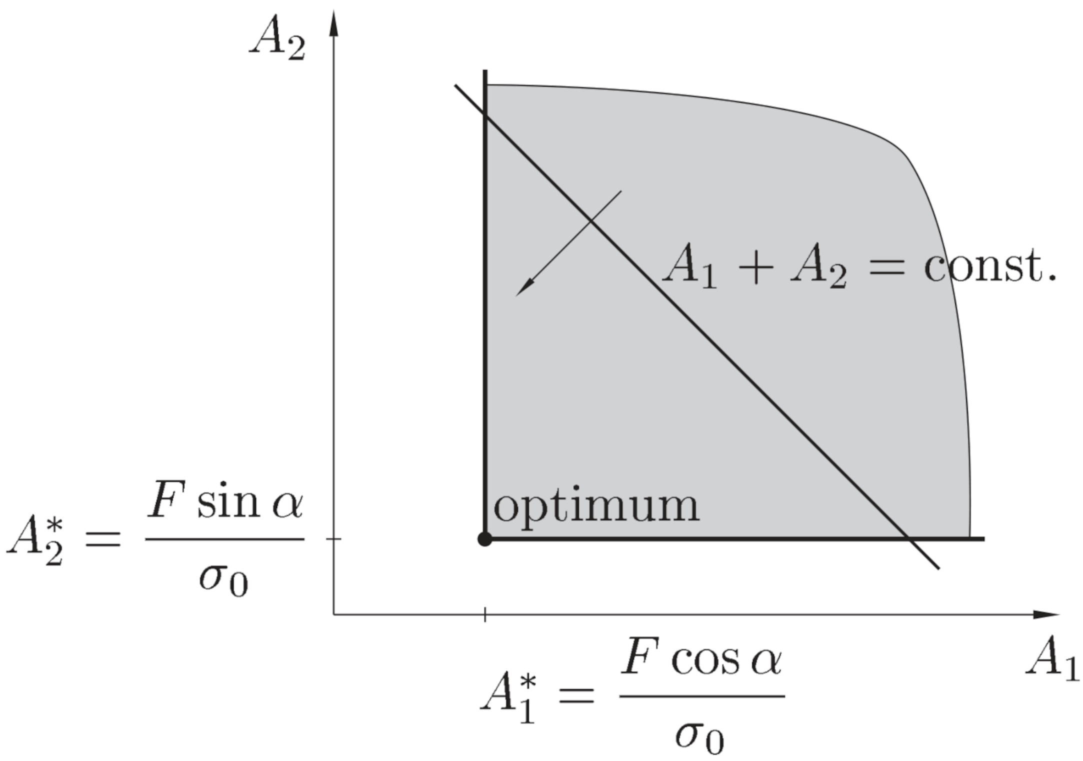
→ 2.1: Two-bar truss s.t. stress constraints

figure → convex (not strictly convex).



→ 2.4 : Cantilever beam s.t. displ. constraint:

$$\min : \; C_1 (x_1 + x_2)$$

$$\text{s.t.} : \; \frac{1}{x_1^3} + \frac{7}{x_2^3} - C_2 \leq 0$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

objective function: linear → convex

Constraint: check Hessian:

$$\begin{bmatrix} \frac{\partial^2 f_1}{\partial x_1^2} & \frac{\partial^2 f_1}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f_1}{\partial x_2 \partial x_1} & \frac{\partial^2 f_1}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} \frac{12}{x_1^5} & 0 \\ 0 & \frac{84}{x_2^5} \end{bmatrix}$$

Sylvester's criterion:

$$\left.\begin{array}{l} \det H_{11} > 0 \text{ in } \mathcal{X} \\ \det \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} > 0 \text{ in } \mathcal{X} \end{array}\right\} \rightarrow \text{pos def}$$

⇒ optimization problem is convex.

3.3 Karush-Kuhn-Tucker (KKT) conditions

– Lagrangian function $\mathcal{L} : \mathbb{R}^n \times \mathbb{R}^l \to \mathbb{R}$ of the optimization problem $(\mathbb{P})$:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = g_0(\mathbf{x}) + \sum_{i=1}^{l} \lambda_i g_i(\mathbf{x})$$

where $\lambda_i, i = 1, \ldots, l$ are called Lagrange multipliers.

3.3 Karush-Kuhn-Tucker (KKT) conditions

– A point $(\mathbf{x}, \boldsymbol{\lambda})$ is a KKT point of $(\mathbb{P})$ if and only if the KKT conditions are satisfied:

$$
\begin{aligned}
\frac{\partial \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})}{\partial x_j} &\leq 0, \qquad \text{if } x_j = x_j^{\max} \\
\frac{\partial \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})}{\partial x_j} &= 0, \qquad \text{if } x_j^{\min} < x_j < x_j^{\max} \\
\frac{\partial \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})}{\partial x_j} &\geq 0, \qquad \text{if } x_j = x_j^{\min} \\
\lambda_i g_i(\mathbf{x}) &= 0 \\
g_i(\mathbf{x}) &\leq 0 \\
\lambda_i &\geq 0 \\
\mathbf{x} &\in \mathcal{X}
\end{aligned}
$$

– For sufficiently regular non-convex problems, the KKT conditions are necessary – but not sufficient – optimality conditions for $(\mathbb{P})$.

– For convex problems, a KKT point is always an optimal point (local and global).

3.3 Karush-Kuhn-Tucker (KKT) conditions

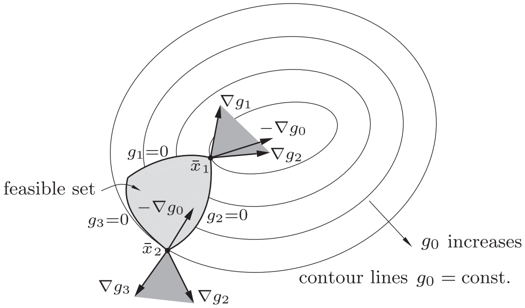
– Geometric interpretation of the KKT conditions when there are no box constraints: $-\nabla g_0(\bar{\mathbf{x}})$ should belong to the cone spanned by the gradients of the active constraints at a point $\bar{\mathbf{x}}$ for $\bar{\mathbf{x}}$ to be a KKT point.

$$\frac{\partial \mathcal{L}}{\partial x_j} = 0$$

$$\Rightarrow \quad \frac{\partial g_0}{\partial x_j} + \sum_{i=1}^{l} \lambda_i \frac{\partial g_i}{\partial x_j} = 0$$

$$\Rightarrow \quad \nabla g_0 + \sum_{i=1}^{l} \lambda_i \nabla g_i = \mathbf{0}$$

$$\Rightarrow \quad -\nabla g_0 = \sum_{i=1}^{l} \lambda_i \nabla g_i$$



$\min \ c_1(x_1+x_2)$
s.t. $\frac{1}{x_1^3} + \frac{7}{x_2^3} \leq c_2$

Example : solution of 2.4 by finding a KKT point.

$$\mathcal{L}(x_1, x_2, \lambda_1) = c_1(x_1+x_2) + \lambda_1\left(\frac{1}{x_1^3} + \frac{7}{x_2^3} - c_2\right)$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = c_1 - 3\frac{\lambda_1}{x_1^4} \begin{cases} \leq 0 & \text{if } x_1 = x_1^{max} \\ = 0 & \text{otherwise} \\ \geq 0 & \text{if } x_1 = x_1^{min} \end{cases} \qquad (1)$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = c_1 - 21\frac{\lambda_1}{x_2^4} \begin{cases} \leq 0 & \text{if } x_2 = x_2^{max} \\ = 0 & \text{otherwise} \\ \geq 0 & \text{if } x_2 = x_2^{min} \end{cases} \qquad (2)$$

$$\lambda_1\left(\frac{1}{x_1^3} + \frac{7}{x_2^3} - c_2\right) = 0 \qquad (3)$$

$$\frac{1}{x_1^3} + \frac{7}{x_2^3} - c_2 \leq 0 \rightarrow x_1 > 0,\ x_2 > 0 \qquad (4)$$

$$\lambda_1 \geq 0 \qquad (5)$$

$$x_1 \geq 0 \qquad (6)$$

$$x_2 \geq 0 \qquad (7)$$

$(4) \rightarrow x_1 > 0, x_2 > 0 \rightarrow (1b), (2b)$

$(1b), (2b) \rightarrow \lambda_1 \neq 0$

$\Rightarrow$ (3) becomes :

$$\frac{1}{x_1^3} + \frac{7}{x_2^3} - c_2 = 0. \qquad (8)$$

Note : it is not always this simple to determine which constraints are active and to solve the resulting system of (non-linear) equations ($\rightarrow$ dual methods)

Equations (1b) and (2b) lead to:

$$x_1 = \left(\frac{3\lambda_1}{c_1}\right)^{1/4} \qquad (9)$$

$$x_2 = \left(\frac{21\lambda_1}{c_1}\right)^{1/4} \qquad (10)$$

Introducing this in (8) leads to:

$$\left(\frac{c_1}{3\lambda_1}\right)^{3/4} + 7\left(\frac{c_1}{21\lambda_1}\right)^{3/4} - c_2 = 0$$

$$\Rightarrow \left(1+7^{1/4}\right)\left(\frac{c_1}{3\lambda_1}\right)^{3/4} - c_2 = 0$$

$$\Rightarrow \left(1+7^{1/4}\right)^{4/3}\frac{c_1}{3\lambda_1} = c_2^{4/3}$$

$$\Rightarrow \lambda_1 = \left(1+7^{1/4}\right)^{4/3}\frac{c_1}{3c_2^{4/3}}$$

Introducing this in (9) and (10) gives:

$$x_1 = \frac{\left(1+7^{1/4}\right)^{1/3}}{c_2^{1/3}}$$

$$x_2 = \frac{7^{1/4}\left(1+7^{1/4}\right)^{1/3}}{c_2^{1/3}}$$

3.4 Lagrangian duality

- The optimization problem $(\mathbb{P})$ can be rewritten as:

$$\min_{\mathbf{x}\in\mathcal{X}} \max_{\boldsymbol{\lambda}\geq 0} \mathcal{L}(\mathbf{x},\boldsymbol{\lambda}) = \min_{\mathbf{x}\in\mathcal{X}} \max_{\boldsymbol{\lambda}\geq 0} \left\{ g_0(\mathbf{x}) + \sum_{i=1}^{l} \lambda_i g_i(\mathbf{x}) \right\}$$

- The (Lagrangian) dual problem $(\mathbb{D})$ corresponding to the primal problem $(\mathbb{P})$ is obtained by interchanging $\min$ and $\max$:

$$\max_{\boldsymbol{\lambda}\geq 0} \min_{\mathbf{x}\in\mathcal{X}} \mathcal{L}(\mathbf{x},\boldsymbol{\lambda}) = \max_{\boldsymbol{\lambda}\geq 0} \min_{\mathbf{x}\in\mathcal{X}} \left\{ g_0(\mathbf{x}) + \sum_{i=1}^{l} \lambda_i g_i(\mathbf{x}) \right\}$$

  or:

$$\max_{\boldsymbol{\lambda}\geq 0} \varphi(\boldsymbol{\lambda})$$

  where the dual objective function φ is defined as:

$$\varphi(\boldsymbol{\lambda}) = \min_{\mathbf{x}\in\mathcal{X}} \mathcal{L}(\mathbf{x},\boldsymbol{\lambda})$$

- For convex problems $(\mathbb{P})$ satisfying Slater's constraint qualification (there exists at least 1 feasible point), the primal problem $(\mathbb{P})$ and the dual problem $(\mathbb{D})$ are equivalent.

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3.4.1 Lagrangian duality for convex and separable problems

– Assume that the objective functions and the constraints are separable functions:

$$g_i(\mathbf{x}) = \sum_{j=1}^{n} g_{ij}(x_j), \quad i = 0, \dots, l$$

– The dual objective function φ is then found by solving n separate one-dimensional optimization problems:

$$\begin{aligned}
\varphi(\boldsymbol{\lambda}) &= \min_{\mathbf{x}\in\mathcal{X}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) \\
&= \min_{\mathbf{x}\in\mathcal{X}} \left\{ g_0(\mathbf{x}) + \sum_{i=1}^{l} \lambda_i g_i(\mathbf{x}) \right\} \\
&= \min_{\mathbf{x}\in\mathcal{X}} \left\{ \sum_{j=1}^{n} g_{0j}(x_j) + \sum_{i=1}^{l} \lambda_i \sum_{j=1}^{n} g_{ij}(x_j) \right\} \\
&= \sum_{j=1}^{n} \min_{x_j\in\mathcal{X}_j} \left\{ g_{0j}(x_j) + \sum_{i=1}^{l} \lambda_i g_i(x_j) \right\}
\end{aligned}$$

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Example : solution of 2.4 using duality

$$\max_{\lambda_1 \geq 0} \varphi(\lambda_1)$$

where $\varphi(\lambda_1) = \min_{\substack{x_1 \geq 0 \\ x_2 \geq 0}} \mathcal{L}(x, \lambda_1)$

$$= \min_{\substack{x_1 \geq 0 \\ x_2 \geq 0}} c_1(x_1 + x_2) + \lambda_1\left(\frac{1}{x_1^3} + \frac{7}{x_2^3} - c_2\right)$$

$$= \underbrace{\min_{x_1 \geq 0} \underbrace{\left(c_1 x_1 + \frac{\lambda_1}{x_1^3}\right)}_{\mathcal{L}_1}}_{\varphi_1} + \underbrace{\min_{x_2 \geq 0} \underbrace{\left(c_1 x_2 + \frac{7\lambda_1}{x_2^3}\right)}_{\mathcal{L}_2}}_{\varphi_2} - c_2 \lambda_1$$

Two one-dimensional opt. problems are solved separately :

$$\frac{\partial \mathcal{L}_1}{\partial x_1} = c_1 - 3\frac{\lambda_1}{x_1^4} = 0 \Rightarrow x_1 = \left(\frac{3\lambda_1}{c_1}\right)^{1/4} \quad (1)$$

$$\Rightarrow \varphi_1(\lambda_1) = c_1\left(\frac{3\lambda_1}{c_1}\right)^{1/4} + \lambda_1\left(\frac{c_1}{3\lambda_1}\right)^{3/4}$$

$$= \frac{4}{3^{3/4}} c_1^{3/4} \lambda_1^{1/4}$$

$$\frac{\partial \mathcal{L}_2}{\partial x_2} = c_1 - 21\frac{\lambda_1}{x_2^4} = 0 \Rightarrow x_2 = \left(\frac{21\lambda_1}{c_1}\right)^{1/4} \quad (2)$$

$$\Rightarrow \varphi_2(\lambda_1) = c_1\left(\frac{21\lambda_1}{c_1}\right)^{1/4} + 7\lambda_1\left(\frac{c_1}{21\lambda_1}\right)^{3/4}$$

$$= 7^{1/4} \frac{4}{3^{3/4}} c_1^{3/4} \lambda_1^{1/4}$$

Dual objective function :

$$\varphi(\lambda_1) = \left(1+7^{1/4}\right) \frac{4}{3^{3/4}} c_1^{3/4} \lambda_1^{1/4} - c_2 \lambda_1$$

Solution of the dual problem :

$$\frac{\partial \varphi}{\partial \lambda_1} = \frac{1}{\cancel{4}}\left(1+7^{1/4}\right) \frac{\cancel{4}}{3^{3/4}} c_1^{3/4} \frac{1}{\lambda_1^{3/4}} - c_2 = 0$$

$$\Rightarrow \lambda_1^{3/4} = \frac{\left(1+7^{1/4}\right) c_1^{3/4}}{3^{3/4} c_2}$$

$$\Rightarrow \lambda_1 = \frac{\left(1+7^{1/4}\right)^{4/3} c_1}{3 c_2^{4/3}}$$

Introducing this in (1) and (2) gives :

$$x_1 = \frac{\left(1+7^{1/4}\right)^{1/3}}{c_2^{1/3}}$$

$$x_2 = \frac{7^{1/4}\left(1+7^{1/4}\right)^{1/3}}{c_2^{1/3}}$$